Project Report: Active AI Surveillance & Deterrent System

1. Project Overview

This project, titled **SentinAI-Cam**, is an autonomous security system that utilizes an ESP32-CAM for wireless CCTV streaming and the Google Gemini 2.5 Flash model for real-time behavioral analysis. Unlike traditional security systems that only record footage, this system implements **Active Defense** by triggering hardware deterrents based on the perceived threat level.

2. System Architecture

The system consists of three primary layers:

- **Acquisition Layer:** An ESP32-CAM captures images from a retail or high-security environment and serves them over a local network.
- **Intelligence Layer (Gemini AI):** A Python-based workstation fetches the frames and sends them to the Gemini API. The AI analyzes the scene for "is_theft", "confidence", and "danger_level" (Low vs. High).
- **Execution Layer (Arduino/Hardware):** An Arduino connected via Serial (COM10) triggers a Buzzer for low-danger theft (shoplifting) or a high-voltage Pulse + Buzzer for high-danger robbery.

3. Threat Logic Table

Observed Behavior	AI Danger Level	Confidence	Action Taken	Hardware Output
Normal Shopping	LOW	High	None	None
Shoplifting/Concealment	LOW	>65%	Warning	Buzzer (b)
Armed Robbery/Masked Entry	HIGH	>85%	Neutralization	Pulse + Buzz (t+b)

4. Key Features

- **Behavioral Analysis:** Moves beyond simple motion detection to understand *intent* (e.g., concealing a bottle).
- **Exponential Backoff:** Ensures system stability even during network fluctuations or API rate limiting.

- **Automatic Logging:** Every detection event is saved as a structured JSON file for forensic evidence.
- **Wireless Flexibility:** The ESP32-CAM allows the camera to be placed anywhere within Wi-Fi range.

5. Technical Stack

- **Hardware:** ESP32-CAM, Arduino Uno, Serial Interface, Deterrent Circuitry.
- **Software:** Python 3.x, `pyserial` , `requests` .
- **AI Model:** Gemini 2.5 Flash Preview.

6. Conclusion

The **SentinAI-Cam** successfully bridges the gap between passive monitoring and active security. By leveraging the reasoning capabilities of Large Multimodal Models (LMMs), the system reduces false alarms and provides a proportional response to criminal activity.